

Database Performance

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September 05, 2019

DB/TP application characteristics

- Typically < 100 transaction types
- Transaction size has high variance, typically
 - 0-30 disk accesses
 - 10K - 1M CPU instructions
 - 2-20 messages
- Example: airline reservations (around year 2000)
 - 150,000 active displays
 - thousands of disk drives
 - 3000 transactions per second

Performance measurements

- Max transactions-per-second or -per-minute (tps, tpm)
- dollars per tps or dollars per tpm
- TP Performance Council (TPC) set standards <http://www.tpc.org>
- Standard benchmarks: TPC-A,-B,-C,-E ...
- TPC-A & -B models a bank teller system
 - Retired ('89-'95), but interesting
 - TPC-A includes terminal and communications and “think time”
 - TPC-B is database only
 - Database tables: {Accounts, Tellers, Branches, History}

TPC-A/B Benchmark

Start

```
Read messages from terminal (100 bytes)
Read + write account record (random access)
Write history record (sequential access)
Read + write teller record (random access)
Read + write branch record (random access)
Write message to terminal (200 bytes)
```

Commit

- Database has 100K accounts account table must be indexed
- End of history and branch records are bottlenecks

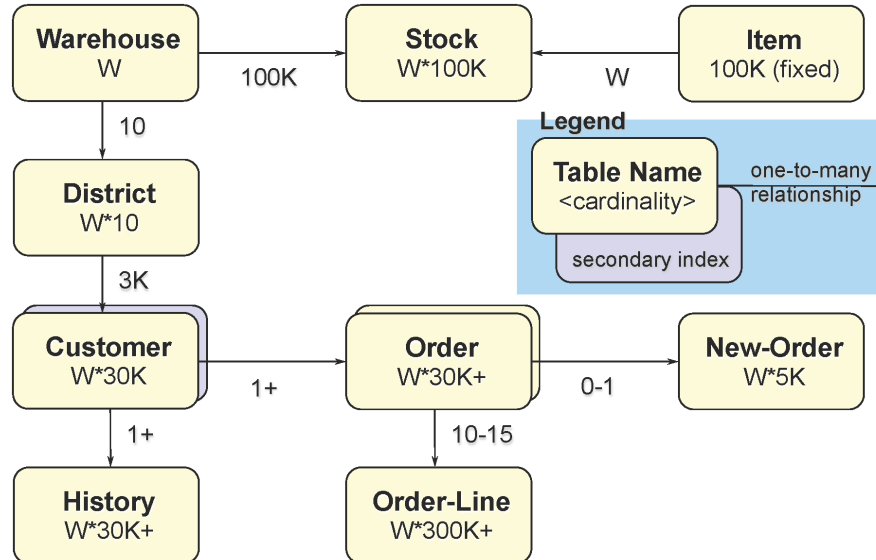
TPC-A/B numbers

- *Huge* effect on products
- Drove down resource consumption to
 - < 100K instructions
 - 2.1 forced disk IOs per transaction
- COO (cost of ownership): Dollars measured by list purchase price plus 5 year vendor maintenance
- Workload profile:
 - 10% TP monitor plus application
 - 30% communications
 - 50% DBMS

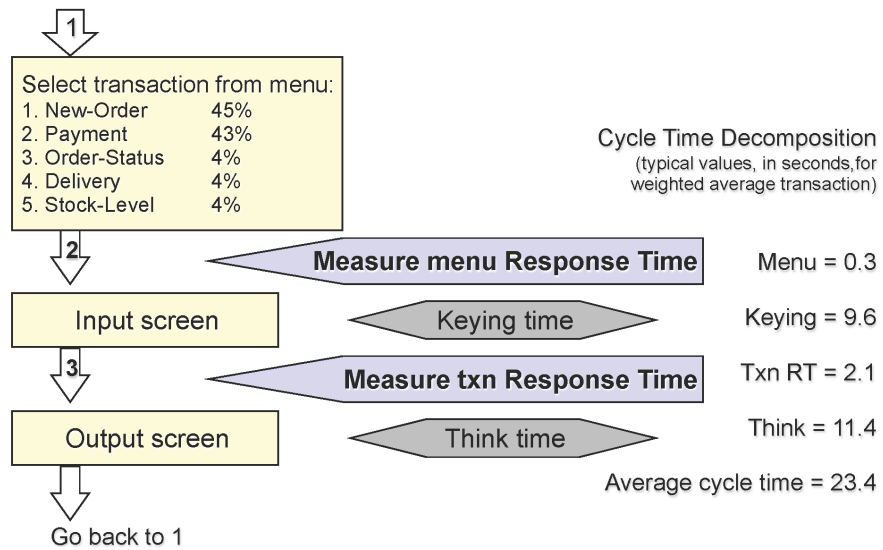
TPC-C benchmark

- Heavier-weight transactions
 - new-order
 - payment, order-status, delivery, stock-level, ... similar complexity, different frequencies
- tpmC = number of new-order tpm

TPC-C database schema



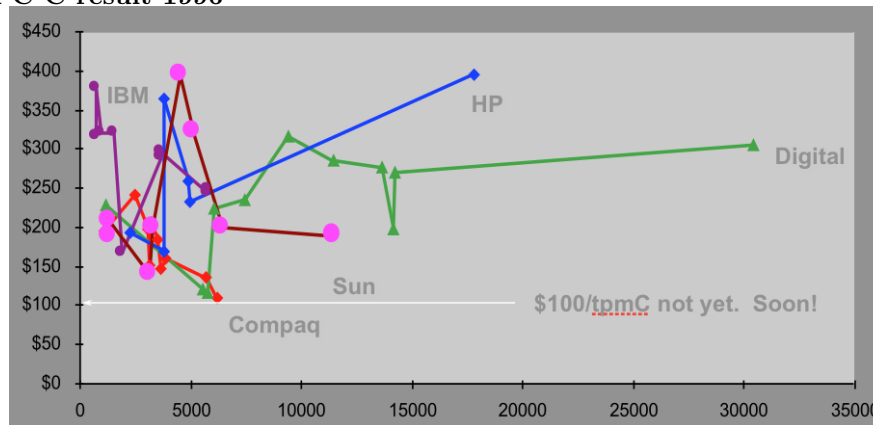
TPC-C workflow



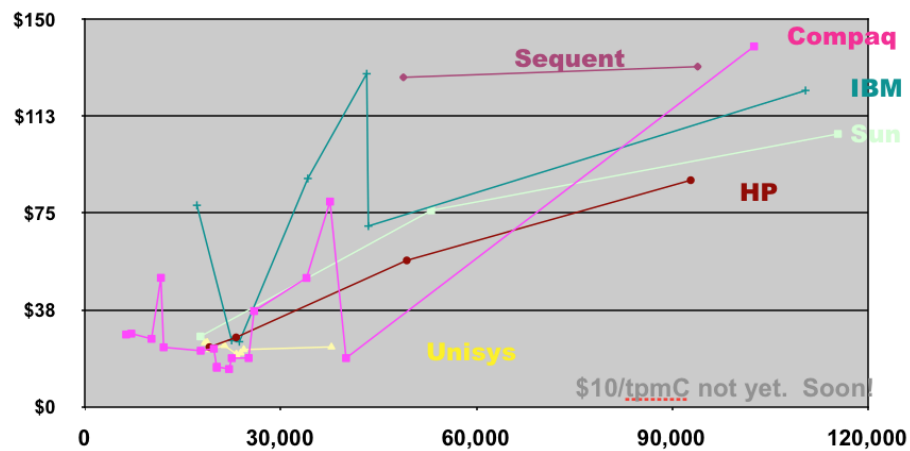
competitive configurations 1996, 1999

1996	1999
5677 tpmC	40,013 tpmC
\$135/tpmC, COO = \$770K	\$19/tpmC, COO = \$755K
2GB RAM	4GB RAM
91 4GB disks	252 9GB + 225 4GB disks
4 Pentium 166MHz	8 Pentium-III 555 MHz
5000 users	32,400 users

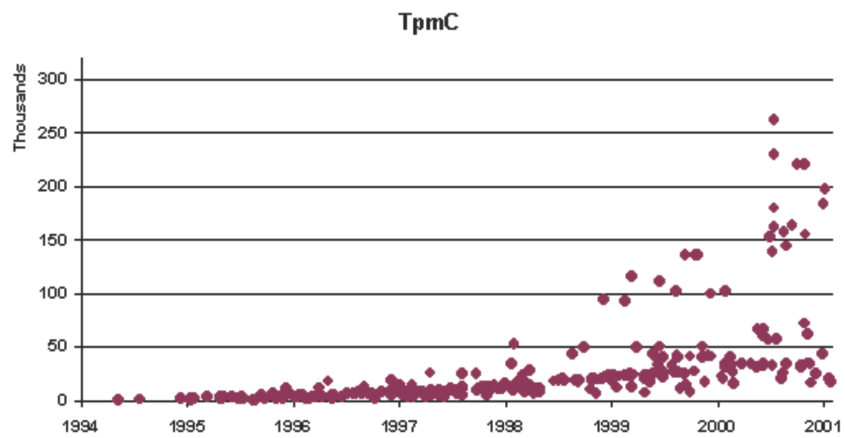
TPC-C result 1996



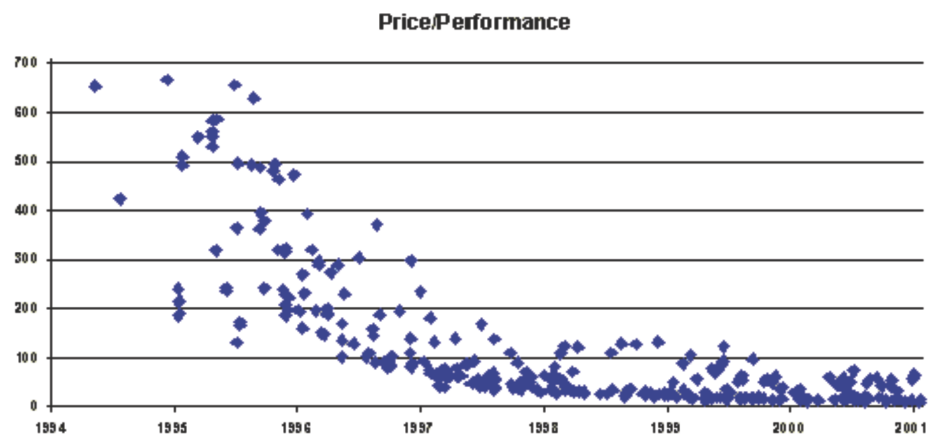
TPC-C result 1999



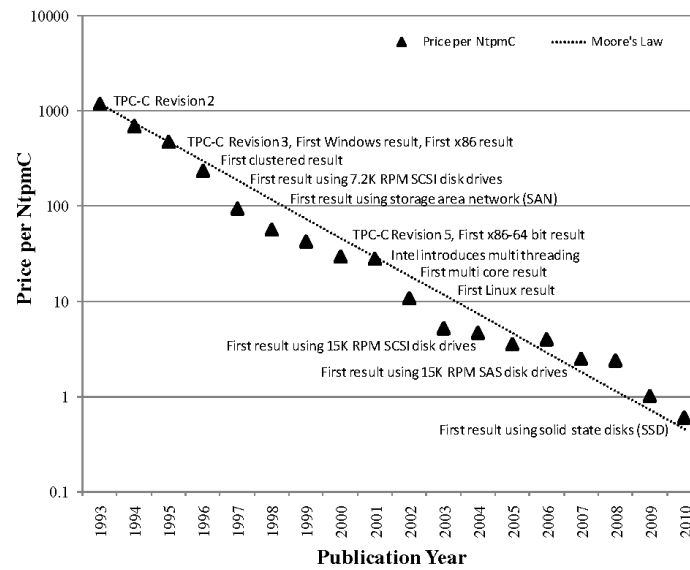
TPC-C performance trends 2001



TPC-C price/performance trends 2001



Moore's law



Price per tpmC normalized with: $f(y) = f(y_0) * 2^{-\frac{2}{3} * (y - y_0)}$

TPC-E benchmark

- started in 2007
- Financial market, brokerage house, C2B, B2B
- tpsE = trade-order tps

characteristics	TPC-C	TPC-E
tables	9	33
foreign keys	9	50
check constraints	0	22
integrity constraints	no	yes